PROGRAM CODE:

```
clc;
s=%s;
z=0;//Undamped system
num=36; den=36+2*z*s+s^2;
TF=syslin('c',num,den);
subplot(221)
t=0:0.1:10;
y1=csim('step',t,TF);
plot(t,y1)
xlabel('Time {t} sec')
ylabel('System response {c(t)}')
title('Undamped system');
z=0.5;//Underdamped system
num=36; den=36+2*z*s+s^2;
TF=syslin('c',num,den);
subplot(222)
t=0:0.1:10;
y2=csim('step',t,TF);
plot(t,y2)
xlabel('Time {t} sec')
ylabel('System response {c(t)}')
title('Underdamped system');

z=1;//Critically damped system
num=36; den=36+2*z*s+s^2;
TF=syslin('c',num,den);
subplot(223)
```

```
t=0:0.1:10;
y3=csim('step',t,TF);
xlabel('Time {t} sec')
ylabel('System response {c(t)}')
plot(t,y1)
title('Critically damped system');
z=10;//Over damped system
num=36; den=36+2*z*s+s^2;
TF=syslin('c',num,den);
subplot(224)
t=0:0.1:10;
y4=csim('step',t,TF);
plot(t,y4)
xlabel('Time {t} sec')
ylabel('System response {c(t)}')
title('Over damped system');
```

OUTPUT:

CS EXPERIMENT 4.sci (C:\Users\PRIYANKA\Documents\CS EXPERIMENT 4.sci) - SciNotes

File Edit Format Options Window Execute ?

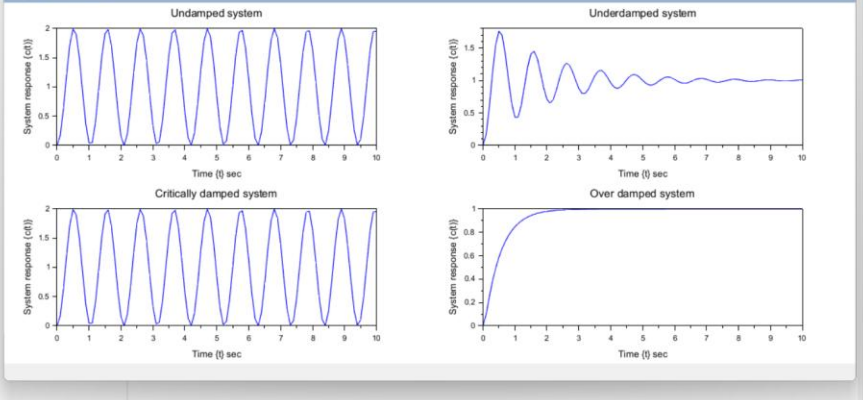
CS EXPERIMENT 2.1.sci | Untitled 1 | CS EXPERIMENT 2.2.sci | CS EXPERIMENT 3.sci | CS EXPERIMENT 4.sci

```
subplot(221)
t=0:0.1:10;
y1=csim('step',t,TF);
plot(t,y1)
xlabel('Time (t) sec')
ylabel('System response (c(t))')
title('Undamped system');
z=0.5; //Underdamped system
num=36; den=36+2*z*s+s^2;
TF=syslin('c',num,den);
subplot(222)
t=0:0.1:10;
y2=csim('step',t,TF);
plot(t,y2)
xlabel('Time (t) sec')
ylabel('System response (c(t))')
title('Underdamped system');

z=1;//Critically damped system
num=36; den=36+2*z*s+s^2;
TF=syslin('c',num,den);
subplot(223)
t=0:0.1:10;
y3=csim('step',t,TF);
xlabel('Time (t) sec')
ylabel('System response (c(t))')
plot(t,y1)
title('Critically damped system');
z=10;//Over damped system
num=36; den=36+2*z*s+s^2;
TF=syslin('c',num,den);
subplot(224)
t=0:0.1:10;
y4=csim('step',t,TF);
plot(t,y4)
xlabel('Time (t) sec')
ylabel('System response (c(t))')
title('Over damped system');
```

Graphic window number 0

File Tools Edit ?



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